Legal Services: Problem Statements

LS-01: The Growth Trap (Compliance Triggered by Business Milestones)

Context:

An SME's legal obligations are not fixed. They switch on silently as the business grows. Hiring the 10th employee can bring in ESI coverage and the requirement for an Internal Committee under the POSH Act. Reaching 20 employees can bring EPF. Crossing a turnover limit triggers GST registration or e-invoicing. Opening a branch in another state brings a fresh set of state-specific registrations. Most owners only find out when a notice or penalty arrives.

The Challenge:

Design a solution that watches for growth events in a small business, such as new hires, rising turnover, new locations, new product lines or a first export order. For each event, it should tell the owner which new obligations have just switched on, by when they must act, and what the consequences of missing them are.

LS-02: What Changed and Does It Affect Me? (Regulatory Change Impact)

Context:

Regulations reach SMEs as gazette notifications, circulars, amendments and FAQs, often dozens a month across central and state authorities. Large companies pay legal teams to track these. Small businesses rely on word of mouth from their accountant, often too late. The underlying problem is that law is published as text, not as a list of changes, and nobody connects a change to a specific business.

The Challenge:

Build a solution that takes a regulatory update and answers three questions for a given SME profile. What exactly changed compared with the earlier version? Does it apply to this business? If so, what must the business do differently, and by when?

LS-03: Paid on Time, by Law (Enforcing MSME Payment Rights)

Context:

Delayed payment is one of the biggest threats to micro and small enterprises. Indian law already protects them. The MSMED Act requires buyers to pay within an agreed period of up to 45 days. Late payments attract compound interest at three times the RBI bank rate. Sellers can approach the MSME Facilitation Council. Since 2023, buyers also lose the income tax deduction on overdue

payments to micro and small suppliers under Section 43B(h). In practice, very few small suppliers use these rights. They don't know how to calculate what is owed, they lack organized documentation, and above all they fear losing the customer.

The Challenge:

Build a solution that helps a micro or small supplier understand, quantify and enforce its payment rights. The approach should also preserve the business relationship wherever possible.

LS-04: Personal Data in Informal Systems (Data Protection for Small Businesses)

Context:

India's Digital Personal Data Protection Act, 2023 applies to businesses of every size that handle personal data digitally, and its rules are being phased in. A typical SME stores customer, employee and patient data across WhatsApp chats, Google Forms, Excel sheets, billing software, CCTV systems and personal phones. Most could not tell you what personal data they hold, where it sits, why they collected it, or how they would respond if a customer asked for it to be erased. They also could not say what they would do in a breach.

The Challenge:

Build a solution that helps a small business reach practical, demonstrable data protection compliance without a dedicated privacy team. That means finding out what personal data it handles, managing consent and purpose, responding to individuals' rights requests, and being prepared for a breach.

LS-05: Notice on the Desk (Government and Legal Notice Triage)

Context:

Small businesses regularly receive notices: GST show-cause notices, income tax intimations, labour department queries, cheque-bounce demand notices, and legal notices from customers, landlords or ex-employees. Many arrive by email, post or a portal alert that nobody checks. They are written in dense legal language, response windows can be as short as 15 to 30 days, and ignoring one can turn a small issue into an ex-parte order or penalty. Owners often can't tell a routine intimation from a serious demand.

The Challenge:

Build a solution that lets an owner upload or forward any notice and tells them, in plain language, who sent it, what law it invokes, what they are being accused of or asked to do, how serious it is, and the exact deadline. It should then guide the response: which documents to gather, whether it needs a professional, and a draft reply structure. It should also track open notices so none slip past their deadline.

LS-06: From Handshake to Contract (Informal Agreements)

Context:

A large share of small business dealings happen informally: a verbal commitment, a WhatsApp conversation confirming price and delivery, a voice note agreeing to a revised deadline. When a dispute arises, the business has no clean agreement to point to. Chats are scattered and ambiguous, and courts expect proper documentation and, in many cases, stamp duty paid under state rules. Owners avoid formal contracts because they seem expensive, slow and awkward to ask for from a long-standing customer.

The Challenge:

Build a solution that helps a small business turn informal dealings into a clear, defensible record without making the relationship feel legalistic. It should extract the key terms from conversations (parties, scope, price, timelines, payment terms), flag anything ambiguous or missing, generate a simple agreement in plain language for both sides to confirm, and preserve the evidence trail in a tamper-evident way.

Finance: Problem Statements

F-01. Financial Fraud and Transaction Risk Detection

FINANCIAL SECURITY
PATTERN ANALYSIS

Financial transactions can exhibit unusual patterns for many reasons. A large payment may be legitimate, while a sequence of individually ordinary transactions may collectively indicate suspicious activity. Financial institutions and payment service providers must identify potential risks without overwhelming investigators with unnecessary alerts.

Problem Statement

Develop a solution that analyses financial transaction activity to identify, assess, and prioritise transactions or sequences of transactions that may warrant further investigation.

The solution should consider relevant factors such as transaction amounts, frequency, timing, locations, transaction history, relationships between accounts, and changes in spending or transfer behaviour. It should account for the possibility that legitimate users may exhibit unusual behaviour and that suspicious activity may not be apparent from individual transactions alone.

The solution should help users understand why an activity was flagged, distinguish different levels of concern, and investigate relationships between potentially connected events. Teams may also consider how the system adapts to changing behaviour, incomplete information, and previously unseen patterns.

Expected Outcome

A functional prototype that identifies and prioritises potentially suspicious financial activity, provides interpretable reasons and supporting transaction context, and enables users to investigate cases while balancing detection effectiveness with unnecessary alerts.

F-02. Multi-Goal Financial Planning Under Uncertainty

PERSONAL FINANCE
DECISION SUPPORT

Individuals often have several financial goals at the same time, such as pursuing higher education, purchasing a home, preparing for retirement, repaying loans, or building an emergency fund. These goals compete for limited resources, and their costs, timelines, and priorities may change over time.

Problem Statement

Develop a solution that helps individuals create, evaluate, and adapt a financial plan across multiple short-term and long-term goals.

The solution should account for income, expenses, existing savings, outstanding liabilities, inflation, changing financial commitments, and the time horizons associated with different goals.

It should help users understand trade-offs when their available resources are insufficient to meet every objective simultaneously.

Users should be able to modify assumptions, introduce unexpected expenses or income changes, reprioritise goals, and compare alternative plans. The solution should communicate the consequences of these decisions, including changes to savings requirements, target dates, and the feasibility of achieving individual goals.

Expected Outcome

An interactive financial planning solution that helps users evaluate competing goals, explore alternative financial scenarios, identify potential shortfalls, and understand the consequences of their decisions over time.

F-03. Predictive Cash-Flow Management for Small Businesses

BUSINESS FINANCE
FORECASTING

A business may be profitable on paper but still struggle to meet its financial obligations because customer payments arrive late, expenses become due unexpectedly, or cash inflows and outflows occur at different times. Small businesses often lack the tools to anticipate these situations and evaluate their options before a shortage occurs.

Problem Statement

Develop a solution that enables small-business owners to understand their current liquidity position, anticipate future cash-flow shortages, and evaluate possible measures to maintain financial stability.

The solution should account for expected revenue, payment schedules, outstanding receivables, supplier obligations, payroll, loan instalments, operating expenses, and available cash reserves. It should distinguish between expected income and money that has actually been received, while accounting for uncertainty in payment timing and business activity.

Users should be able to simulate scenarios such as delayed customer payments, unexpected expenses, reduced sales, changes in supplier terms, or revised payment schedules. The solution should help them identify when a shortage may occur and compare possible responses based on their financial implications.

Expected Outcome

A practical cash-flow management system that presents the current and projected cash position, identifies periods of potential financial stress, and enables business owners to compare scenarios and evaluate appropriate corrective actions.

F-04. Fair and Explainable Creditworthiness Assessment

LENDING

FINANCIAL INCLUSION

Access to credit is important for individuals and small businesses, but assessing the ability to repay a loan can be difficult when financial records are limited, inconsistent, or incomplete. Conventional assessment approaches may not adequately capture a borrower's actual repayment capacity, while inaccurate assessments can expose lenders and borrowers to financial risk.

Problem Statement

Develop a solution that helps evaluate the creditworthiness of individuals or small businesses using relevant financial information and clearly communicates the factors influencing the assessment.

The solution should consider factors such as income stability, existing obligations, repayment history, cash-flow patterns, debt burden, and the reliability or completeness of available information. It should distinguish between an absence of financial history and evidence of an inability to repay.

The solution should help users understand the strengths, limitations, and uncertainties of an assessment. Teams may explore how missing information, changing financial circumstances, different loan terms, or alternative repayment scenarios affect the resulting assessment. The approach should also consider consistency, fairness, and the risk of drawing conclusions from incomplete or misleading data.

Expected Outcome

A working credit-assessment prototype that produces an interpretable evaluation of repayment capacity, identifies relevant risk factors and information gaps, and demonstrates how different financial circumstances or loan conditions affect the assessment.

F-05. Financial Visibility and Hidden Expense Discovery

CONSUMER FINANCE

SPENDING ANALYSIS

Individuals often struggle to understand where their money goes, particularly when spending is distributed across multiple accounts, payment methods, recurring subscriptions, loan repayments, and everyday purchases. Individually small expenses can accumulate into significant financial commitments, while irregular spending and poorly understood payment patterns can make budgeting difficult.

Problem Statement

Develop a solution that transforms fragmented financial activity into a clear, useful picture of an individual's spending behaviour, recurring commitments, and overall financial position.

The solution should help users categorise transactions, identify recurring payments, distinguish essential commitments from discretionary spending, and recognise meaningful changes in their financial behaviour. It should account for transfers between a user's own accounts, refunds,

irregular income, shared expenses, and other transactions that may distort a simple spending summary.

Users should be able to investigate unexpected spending, estimate the future impact of recurring commitments, compare actual expenditure against a budget, and explore practical adjustments to improve their financial position. The solution should make its conclusions understandable and allow users to correct inaccurate classifications or assumptions.

Expected Outcome

A financial visibility tool that consolidates transaction information into an understandable spending picture, identifies recurring or overlooked expenses, highlights emerging financial pressures, and helps users evaluate actionable ways to manage their money.

F-06. Investment Portfolio Risk and Scenario Analysis

INVESTMENTS
RISK ANALYSIS

Investors must often make decisions across multiple assets with different levels of risk, expected returns, liquidity, and time horizons. A portfolio that appears diversified may still be exposed to common market risks, while a change in one investment can affect the overall portfolio in unexpected ways. Understanding these interactions is difficult when decisions are based on isolated assets or historical returns alone.

Problem Statement

Develop a solution that helps investors understand the composition, performance, and risk exposures of an investment portfolio and evaluate how different market conditions could affect its value.

The solution should consider factors such as asset allocation, historical performance, volatility, concentration, liquidity requirements, investment horizons, and relationships between assets. It should help users examine how portfolio behaviour may change under scenarios such as market declines, interest-rate changes, sector-specific disruptions, or fluctuations in currency values, where relevant to the selected assets.

Users should be able to compare alternative portfolio allocations or investment scenarios, understand the assumptions behind the analysis, and identify potential conflicts between their financial objectives, risk tolerance, and liquidity needs. The solution should distinguish historical observations from hypothetical outcomes and avoid presenting uncertain projections as guaranteed results.

Expected Outcome

An interactive portfolio analysis solution that presents portfolio composition and relevant risk exposures, compares hypothetical market scenarios, and helps users understand the potential consequences and trade-offs of alternative investment decisions.